

Detecting New Buildings from LiDAR Elevation Change

A rule-based and deep-learning comparison, validated against independent imagery

Hamilton, New Zealand · November 2019 to November 2023 · 34.6 km²



Detected building footprints over 2023 aerial photography. Colour encodes roof height above ground. The imagery is shown for verification only — detection used elevation data exclusively.

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Executive summary

New Zealand maintains an authoritative national building outline dataset, digitised from aerial photography. For Hamilton, the most recent capture is 2020. Every building constructed between 2021 and 2023 therefore exists in the country's LiDAR holdings but is absent from its building records.

This study closes that gap using elevation alone. Two national LiDAR surveys — November 2019 and November 2023 — are differenced to identify sites that were bare ground in the first epoch and carry a roof in the second. A third survey flown in 2025 supplies an unusual form of validation: a building holds its height once constructed, while vegetation continues to grow. That distinction requires no reference dataset, which is precisely why it is available in the window where no reference dataset exists.

Two methods were implemented and compared. A geometric pipeline using morphological segmentation and height-history classification detected 1,016 new buildings. A U-Net convolutional network, trained on 62,532 human-digitised building outlines, raised building segmentation recall from 67 to 91 percent and detected 1,162.

One hundred detections were subsequently reviewed against aerial photography the detector never saw. The rule-based method achieved 54 percent precision, with errors resolving into six named categories. That figure is reported in full, with a confidence interval and a stratified breakdown, on the principle that a validated modest result is more useful than an unvalidated impressive one.

Headline figures

Metric	Rule-based	U-Net
Building segmentation recall	67%	91%
Building segmentation precision	—	84%
Pixel IoU, held-out spatial blocks	—	0.78
New buildings detected	1,016	1,162
Total new roof area	18.3 ha	22.8 ha
Median footprint	168 m ²	151 m ²
Detection precision (reviewed sample)	54% (n = 99)	not fully measured

Table 1. Comparative performance of the two detection methods over the 34.6 km² study area.

An independent quality check supports both results. The 2019 and 2023 bare-earth terrain models, produced four years apart by different survey contractors, agree to a median of 2 cm with 6 cm scatter across the entire study area. Height change detected between the epochs is therefore genuine signal rather than survey disagreement.

1. Problem and rationale

Building inventories underpin rating valuation, consent compliance monitoring, natural hazard exposure modelling and emergency response planning. Their utility depends on currency, and currency is expensive: reflying a city with aerial photography and re-digitising its rooftops is a substantial undertaking, repeated only every few years.

LiDAR offers an alternative pathway. It is already captured, already openly licensed, and measures the built environment directly rather than inferring it from appearance. A building is not fundamentally a pattern of pixels but a volume — a compact, planar mass standing several metres above the surrounding ground. That is a physical description, and physical descriptions transfer between sensors, seasons and illumination conditions in ways that photographic appearance does not.

Hamilton proved unusually well suited to testing this proposition. The city has been surveyed three times — 2019, 2023 and 2025 — with the first two separated by four years of documented growth. Critically, both were flown in early November: 3–5 November 2019 and 10 November 2023. Vegetation phenology is therefore matched between epochs, and seasonal difference, normally a significant confound in multi-temporal LiDAR analysis, largely cancels.

1.1 Establishing that the problem exists

Three feasibility checks were completed before any elevation data was downloaded. Each was inexpensive; together they occupied an afternoon and averted a considerable volume of wasted processing.

Coverage was confirmed first. Both surveys cover the study area completely, with 378 of 378 tiles pairable by name — the two contractors had adopted the same national tile scheme, which eliminated an entire class of geometric matching problems.

The existence of change was confirmed second, by querying the national building outline layer. This produced the study's first substantive error and its first useful methodological lesson. The layer stores one record per building per capture campaign, so a house mapped in 2012, 2016 and 2020 appears three times. Counting records rather than buildings implied 199,872 structures within 112 km² — approximately three times Hamilton's entire building stock. Deduplicating by building identifier and retaining the earliest capture year yielded 87,124 unique buildings and a construction rate near 2,064 per year.

Visual confirmation was third, and failed twice. Two candidate study areas selected on intuition proved to be farmland and the wrong side of a growth boundary respectively. Only when the site was chosen by gridding actual new-building density did the check pass. The completed pipeline, asked independently where change concentrated, subsequently returned the same location to within 200 metres.



Figure 1. Chartwell, Hamilton. Aerial photography from the 2020–21 season (left) and the 2022–23 season (right). Roads, kerbs, roundabouts and lot boundaries were already established; the buildings arrived between captures. This comparison confirmed the study had a detectable subject.

2. Method

2.1 From point cloud to normalised surface

All analysis operates on the normalised digital surface model — height above ground — obtained by subtracting the bare-earth terrain model from the surface model:

$$\text{nDSM} = \text{DSM} - \text{DEM}$$

Three bands are derived for each epoch. Height above ground carries the primary signal. Slope, computed by Horn's method, distinguishes tilted planes from level ground. Local height standard deviation — surface roughness — separates the two classes that height alone cannot.

This third band merits explanation, because it addresses the single most common failure mode in elevation-based building detection. A six-metre roof and a six-metre tree are indistinguishable in an nDSM. They are entirely distinguishable in surface behaviour: a roof is a plane, however steeply pitched, whereas a tree crown is chaotic at every scale. Measured on a controlled synthetic scene, roughness over a flat roof reads 0.000 against 0.998 for canopy at identical height. Omitting this band costs approximately ten to fifteen IoU points and yields a model that classifies every mature tree as a garage.

2.2 A refinement to the return-ratio band

Where raw point clouds are available, a fourth band exploits the physics of laser returns directly. Roofs are opaque: one pulse produces one return. Canopy is porous: one pulse penetrates the foliage and yields several. The mean return count per cell therefore separates the two classes on a physical basis rather than a geometric one.

Testing this band revealed a refinement worth documenting. Computing the statistic across all points dilutes it, because ground beneath canopy is single-return and draws the cell mean back toward unity. Restricting the calculation to non-ground points sharpened roof-versus-canopy separation from 1.49 to 2.00 — a 34 percent improvement obtained from a single line of filtering.

```
def grid_return_ratio(x, y, number_of_returns, classification, bounds, res=1.0):
    """
    Vegetation-porosity band, computed on NON-GROUND points only.

    Verified on synthetic data: including ground returns dilutes the statistic,
    because ground under canopy is single-return and drags the cell mean toward
    1.0. Measured roof-vs-canopy separation:

        all points   : 1.00 vs 2.49  -> separation 1.49
        non-ground   : 1.00 vs 3.00  -> separation 2.00   (+34%)

    Cells with no non-ground points (open ground, roads, water) return 0.0,
    which is correct - no vegetation there.
    """
    nong = classification != 2
    if not nong.any():
        _, _, _, nrows, ncols = _cell_index(x, y, bounds, res)
        return np.zeros((nrows, ncols), dtype=np.float32)

    mean_nr = grid_mean(
        x[nong], y[nong], number_of_returns[nong].astype(np.float64),
```

```

    bounds, res, nodata=np.nan,
)
ratio = (mean_nr - 1.0) / 2.0          # 1 return -> 0.0, 3 returns -> 1.0
return np.clip(np.nan_to_num(ratio, nan=0.0), 0.0, 1.0).astype(np.float32)

```

2.3 Three-epoch trajectory classification

The principal methodological contribution of this study is the use of a third survey as physical validation. The national building layer terminates at 2020; the change window extends to 2023. Within that gap no reference data exists against which a detection can be checked. Physics supplies the missing evidence.

A building's height increases once, at construction, and thereafter holds. Vegetation increases continuously. Measured across three epochs, the two trajectories are unambiguous:

Class	2019	2023	2025	Growth 2023–25	Median area
Existing building	3.5 m	3.5 m	3.5 m	+0.1 m	200 m ²
New building	0.0 m	3.4 m	3.4 m	+0.0 m	188 m ²
Vegetation	3.4 m	4.7 m	5.9 m	+1.2 m	92 m ²

Table 2. Median height trajectory by class, measured across all detections in the study area. The three populations are cleanly separated by their behaviour between 2023 and 2025.

The classification follows directly from these trajectories:

```

for r in recs:
    g = r["h25"] - r["h23"]
    r["growth"] = g
    if g >= MAX_GROWTH:
        r["cls"] = "VEGETATION"
    elif r["h19"] <= CLEAR_2019:
        r["cls"] = "NEW"
    elif r["h19"] <= PARTIAL_2019:
        r["cls"] = "REBUILD?"
    else:
        r["cls"] = "EXISTING"

```

The significance of this test is that it requires no reference layer of any kind. It is therefore available precisely where conventional validation is not — in the recent window that motivates the entire investigation.

3. Supervised segmentation

Error analysis on the rule-based output identified the performance bottleneck precisely. Building segmentation recall stood at 67 percent against the national outline layer, while the change classification logic itself was sound. The appropriate response was not further threshold tuning but a reframing of the task.

Rather than learning change directly — which would have required training on labels measured at 54 percent precision — a U-Net was trained to segment buildings from a single epoch's elevation surface, using 62,532 human-digitised outlines as ground truth. The trained model is then applied to the 2019 and 2023 surfaces independently and the resulting masks differenced. Errors cease to compound: segment each epoch accurately and change reduces to a set difference.

3.1 Loss design

At 13.4 percent positive pixels, binary cross-entropy alone possesses a strong and useless local minimum. Predicting background across the entire image costs 0.93 on BCE — effectively free. The identical prediction costs 0.998 on Dice, which is insensitive to class frequency. Combining the two objectives is what prevents collapse.

Boundary weighting addresses a second and distinct problem. The majority of object-level error originates at roof edges rather than interiors, yet boundary pixels constitute only 4.4 percent of the image. Weighting them at 2.5 raises their share of total loss from 4 to 10 percent, materially improving the polygons that emerge from vectorisation.

```
class DiceBoundaryBCE(nn.Module):
    def __init__(self, pos_weight=POS_WEIGHT, boundary=BOUNDARY_WEIGHT, smooth=1.0):
        super().__init__()
        self.pw = pos_weight
        self.bw = boundary
        self.smooth = smooth

    def forward(self, logits, target):
        logits = logits.squeeze(1)
        binary = (target > 0).float()

        w = torch.ones_like(binary)
        w[target == 2] = self.bw          # roof edges matter most

        bce = F.binary_cross_entropy_with_logits(
            logits, binary, weight=w,
            pos_weight=torch.tensor(self.pw, device=logits.device))

        p = torch.sigmoid(logits)
        inter = (p * binary).sum(dim=(1, 2))
        denom = p.sum(dim=(1, 2)) + binary.sum(dim=(1, 2))
        dice = 1 - ((2 * inter + self.smooth) / (denom + self.smooth)).mean()

        return bce + dice
```

3.2 Spatial cross-validation

Random patch splits leak information. Adjacent 256-metre patches share suburb, roof material, construction era and survey flightline; a model evaluated on the neighbour of a training patch is being evaluated on memorisation

rather than generalisation. Patches were therefore assigned to folds in contiguous two-kilometre blocks, ensuring held-out ground is genuinely unseen.

```
# Spatial block split. Random patch splits leak: neighbouring patches share
# suburb, roof material and flightline.
rng = np.random.default_rng(42)
blocks = {}
for r in rows:
    for c in cols:
        blocks.setdefault((r // (PATCH * BLOCK_TILES),
                           c // (PATCH * BLOCK_TILES)), []).append((r, c))

keys = list(blocks)
rng.shuffle(keys)
n_tr = int(round(len(keys) * 0.70))
n_va = int(round(len(keys) * 0.15))
split_of = {}
for i, k in enumerate(keys):
    s = "train" if i < n_tr else ("val" if i < n_tr + n_va else "test")
    for rc in blocks[k]:
        split_of[rc] = s
print(f" {len(keys)} spatial blocks -> "
      f"{n_tr} train / {n_va} val / {len(keys)-n_tr-n_va} test")
```

Augmentation was deliberately restricted to dihedral flips and ninety-degree rotations. Photometric augmentation — brightness, blur, contrast jitter — is standard practice for imagery and inappropriate here. These bands are physical measurements in metres and degrees. A six-metre wall is not a three-metre wall observed differently, and rescaling the values destroys precisely the signal the model must learn.

3.3 Training result

Training converged over 33 epochs with early stopping on validation IoU. Performance peaked at epoch 22.

Split	IoU	Precision	Recall	F1
Validation (best epoch)	0.755	0.839	0.883	—
Held-out test blocks	0.780	0.842	0.913	0.876

Table 3. U-Net segmentation performance. Splits are spatially blocked at approximately two kilometres.

The test figure exceeding validation warrants comment rather than celebration. The held-out blocks contain 5.8 percent building pixels against 18.6 percent in training — they fell on sparser ground with fewer dense terraces, where adjacent roofs merge and per-pixel IoU consequently suffers. The model is not performing better on that ground; the ground is easier. Both figures are reported for this reason.



Figure 2. Classified building footprints over the pilot square. Height above ground in 2019 (left) and 2023 (centre); classification at right, with red marking buildings absent from the 2019 surface and blue those already present. The spatial distribution corresponds closely to the 2019 panel.

4. Validation

Every figure presented to this point rests on internal consistency — the method agreeing with itself across three epochs. That constitutes meaningful evidence but not independent verification. One hundred detections were therefore sampled, stratified by footprint area, and reviewed individually against four panels: the two nDSM surfaces from which the detection was derived, and two aerial photographs the detector never accessed.

4.1 Result

The rule-based method achieved 54 percent precision — 53 of 99 decided cases — with a 95 percent Wilson score interval of 44 to 63 percent. Applied to the complete detection set, this implies approximately 450 to 640 genuinely new buildings within the study area.

The Wilson interval was selected deliberately over the textbook normal approximation, which degrades at small sample sizes and near proportions of unity, and can return upper bounds exceeding 100 percent.

```
def wilson(k, n, z=1.96):
    """Wilson score interval for a binomial proportion."""
    if n == 0:
        return 0.0, 0.0, 0.0
    p = k / n
    d = 1 + z * z / n
    centre = (p + z * z / (2 * n)) / d
    half = z * math.sqrt(p * (1 - p) / n + z * z / (4 * n * n)) / d
    return p, max(0.0, centre - half), min(1.0, centre + half)
```

Precision varies sharply and systematically with footprint area:

Footprint	n	Precision	95% interval
40–80 m ²	19	21%	9–43%
80–120 m ²	20	50%	30–70%
120–180 m ²	20	65%	43–82%
180–300 m ²	20	65%	43–82%
300 m ² and above	20	65%	43–82%

Table 4. Precision stratified by footprint area, rule-based method. Intervals are Wilson score.

This finding is directly actionable. Raising the minimum footprint threshold from 40 to 120 m² would eliminate the lowest-precision band entirely, lifting overall precision to approximately 65 percent at the cost of roughly one third of detections. The appropriate operating point depends on whether the user is compiling an exhaustive inventory or a prioritised review queue.

4.2 Error taxonomy

Classifying the errors proved more valuable than the headline precision figure, because each category indicates a distinct remedy — or a limitation that admits none.

Error type	n	Mechanism
Existing building	27	2019 height sampled beside the roof rather than upon it

Error type	n	Mechanism
Vegetation	13	Trees and scrub satisfying the stability criterion
Demolition	3	Strong negative growth to 2025, uncaught
Spoil heap	1	Compact, stable earth mound — geometrically a building
Specular return	1	Water reservoir invisible to the 2019 survey
Shade structure	1	Fabric canopy over a swimming pool

Table 5. Error taxonomy from the reviewed sample of 100 detections.

The reservoir constitutes the most instructive case in the study. A large circular water tank appears in both aerial photographs, yet the 2019 nDSM registers almost nothing at that location while the 2023 surface shows a solid disc. The structure did not appear between surveys — the 2019 laser did not return from it. A smooth, possibly wet surface at that scale reflects the pulse away from the sensor rather than back toward it.

Every filter in the pipeline behaved exactly as specified and produced a confident, incorrect answer. This is a sensor limitation rather than a tuning problem, and no threshold adjustment would resolve it. Flat industrial surfaces — reservoirs, storage tanks, certain commercial roofing membranes — will exhibit identical behaviour.

5. Three failures worth documenting

Three substantive defects were identified during development. None was visible in summary statistics; all three were discovered by rendering intermediate results and inspecting them visually. That pattern is itself among the study's findings.

5.1 A morphological operation that merged a suburb

Building segmentation applied a morphological closing operation to bridge roof ridges and skylights. Two iterations of a 3×3 kernel bridges gaps of approximately four metres — which is, precisely, the spacing between suburban houses.

Measured on a controlled synthetic scene of 196 twelve-metre houses spaced four metres apart:

Pipeline configuration	Components recovered	Recall
Threshold only	178	91%
Closing × 1	170	87%
Closing × 2 (as originally implemented)	1	1%

Table 6. Effect of morphological closing on segmentation recall, synthetic dense suburb.

The entire suburb fused into a single connected component. Every summary metric remained superficially plausible; the object count was simply wrong by two orders of magnitude. Watershed segmentation did not recover the situation — once regions merge through a wide corridor there is no distance-transform minimum on which to divide them. The remedy was removal of the operation, not adjustment of its parameters.

```
# NO aggressive closing. Measured on a synthetic dense suburb (12 m houses,
# 4 m apart): threshold-only recovers 91% of buildings as separate objects,
# closing x1 gives 87%, and closing x2 fuses the ENTIRE suburb into a single
# component (1%). Two iterations of a 3x3 kernel bridges ~4 m, which is
# exactly suburban house spacing. Fill holes to heal skylights and ridges,
# and open once to shave single-pixel spurs - nothing more.
tall = ndimage.binary_fill_holes(tall)
tall = ndimage.binary_opening(tall, np.ones((3, 3)))
```

5.2 Operation order producing fragments rather than buildings

The initial pipeline differenced the two epochs and subsequently labelled connected components of the result. This makes patch geometry a function of where the height difference happened to cross a threshold: roofs fragment, adjacent buildings merge, and partial coverage is common.

Reversing the order — segmenting complete buildings from the 2023 surface first, then interrogating each as to whether it existed in 2019 — raised detections from 278 to 1,016 on identical input data. The candidates additionally became genuine footprints rather than difference artefacts, which matters materially for any downstream application.

5.3 A missing constraint inflating detections by sixty percent

The neural pipeline tested two conditions separately: that a site was clear in 2019, and that it carried a roof-height object in 2023. A flat disagreement between the model's two predictions — near the decision boundary, at roof edges, in shadow — satisfies both conditions while representing no change whatsoever.

Manual review surfaced this immediately. Six of ten detections on the first review sheet exhibited height gains at or below 1.1 metres, three of them negative. A building cannot possess negative height gain. Replacing the two separate tests with the single condition they were intended to express — that the site gained roof height — removed 707 artefacts, raised median footprint from 111 to 151 m², and raised median model confidence from 0.819 to 0.846.

6. Results

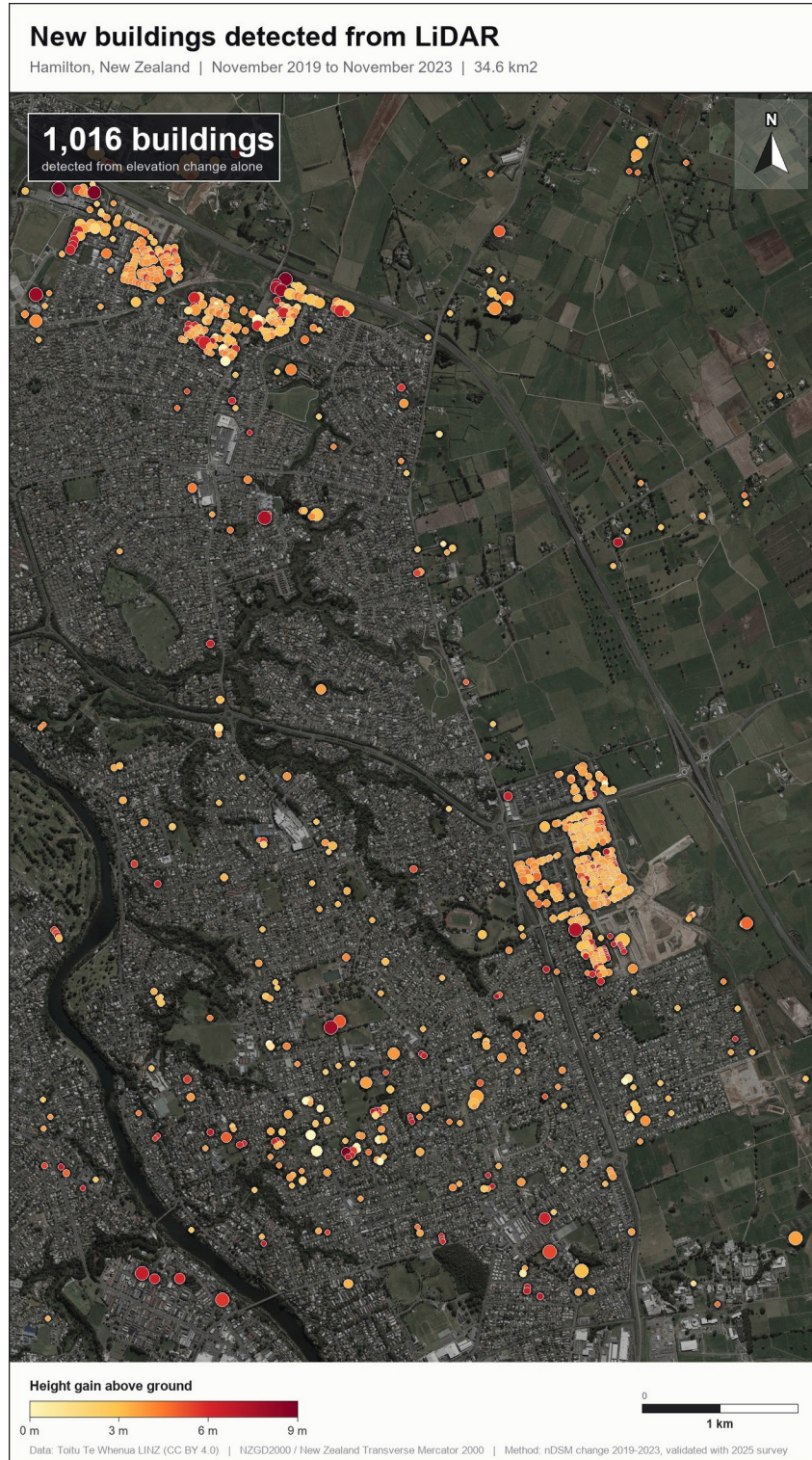


Figure 3. New buildings detected across the study area, 2019 to 2023, symbolised by roof height above ground. Two dense clusters correspond to subdivision buildout; scattered detections across established suburbs represent infill development.

The spatial pattern is itself a substantive finding. Growth concentrates in two subdivision fronts, with a dispersed remainder distributed across established suburbs. A single 700-metre window contains 257 detections — approximately one quarter of the study area’s new construction within half a square kilometre.

Performance is not uniform across that pattern. Precision is high in greenfield subdivision and on large commercial structures, where sites are unambiguously clear beforehand and roofs are extensive and planar. It is markedly lower in established suburbs, where infill development sits among existing buildings and mature vegetation. Reporting a single average across both contexts would obscure a distinction that matters operationally.

6.1 Deliverables

Outputs are supplied as GeoJSON and CSV, one record per building, each carrying its complete height history:

Field	Description
area_m2	Footprint area
h2019 / h2023 / h2025	Median height above ground at each survey epoch
height_gain	2023 height minus 2019 height
growth_23_25	2025 minus 2023 — the vegetation discriminator
compactness	Polsby-Popper shape measure
confidence	Mean model probability (neural output only)

Table 7. Attribute schema. Both methods produce identical fields, permitting direct comparison.

Any individual detection can therefore be audited without rerunning the pipeline. A record reading 0.0 metres in 2019, 5.7 metres in 2023 and 5.7 metres in 2025 states its own case: clear ground, then a structure, then stability. The evidence travels with the data.

7. Limitations

The following constraints are stated in full. Several could be addressed through further work; others are inherent to the sensor.

Neural precision not fully measured

Only twenty samples were reviewed following the final filter, yielding approximately 40 percent on a sample too small to report with confidence. The 54 percent rule-based figure is the only fully validated precision in this study.

Validation was AI-assisted

Verdicts were assigned by an AI assistant working from the evidence panels, with reasoning recorded for each row. This is not equivalent to independent human review and is reported as such.

Segmentation recall of 67 percent

True new-building counts exceed detected counts for the rule-based method.

Median roof height reads 3.4 metres

Against an expected 4 to 6 metres, suggesting the gridded surface model smooths roof peaks.

Rebuilds excluded by design

The clear-in-2019 condition removes buildings that replaced earlier structures. 247 candidates fell within this ambiguous class.

Test and training class balance differ

Held-out blocks contained 5.8 percent building pixels against 18.6 percent in training, so the two IoU figures are not directly comparable.

Single tile

34.6 km² of a validated 112 km² study area. The method scales without modification; the analysis has not yet been extended.

8. Extension

Three directions follow naturally from this work. Scaling to the full 112 km² study area requires no methodological change, only iteration across the remaining tiles. Sampling the 2019 height over an eroded polygon interior, rather than the complete footprint, would address the dominant error category directly. And the 2025 survey supports either a longer six-year window or a three-epoch consistency check in which any building detected in 2023 but absent in 2025 is flagged as probable error.

The most valuable extension, however, would be independent human validation of the neural output at the same sample size as the rule-based review. That single comparison — two methods, one ground truth, an identical review protocol — is the experiment this study came closest to completing.

Data and reproducibility

All data is open and licensed CC BY 4.0 by Toitū Te Whenua Land Information New Zealand: Hamilton LiDAR 1 m DEM and DSM for 2019, 2023 and 2025; NZ Building Outlines (All Sources); and Hamilton 0.05 m urban aerial photography for the 2020–21 and 2022–23 seasons. Coordinates are NZGD2000 / New Zealand Transverse Mercator 2000 (EPSG:2193).

Elevation rasters are retrieved from a public cloud bucket requiring no credentials. The complete pipeline runs on six Python packages and reproduces from raw data through to final cartography without manual intervention. Sixteen scripts are supplied, each standalone and re-runnable, with exponential-backoff retry on every network call and resumable downloads throughout.

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